

YOLO Detection Dataset Spectral Comparison

Particles are exact YOLO label intervals.

Dataset noise is made of FFT windows that do not overlap any label.

Standalone Noise is a single external reference, shown once in neutral color.

Datasets:

- YOLO v3 (old pipeline): P1/yolo_dataset_v3
- C1 Hamming (refined labels): P0/data/dataset_hamming_c1_yolo
- Standalone Noise reference: P0/data/Noise

Coverage summary:

C1 Hamming (refined labels) test: signals=578, labels=6905, mean coverage=66.6%, p95=93.9%

YOLO v3 (old pipeline) test: signals=354, labels=384, mean coverage=7.0%, p95=14.1%

10-40 kHz and overlap summary:

YOLO v3 (old pipeline) 2um: particle=71.0%, dataset-noise=19.1%, Noise reference=19.0%, dominant overlap=False

YOLO v3 (old pipeline) 4um: particle=93.0%, dataset-noise=19.2%, Noise reference=19.0%, dominant overlap=False

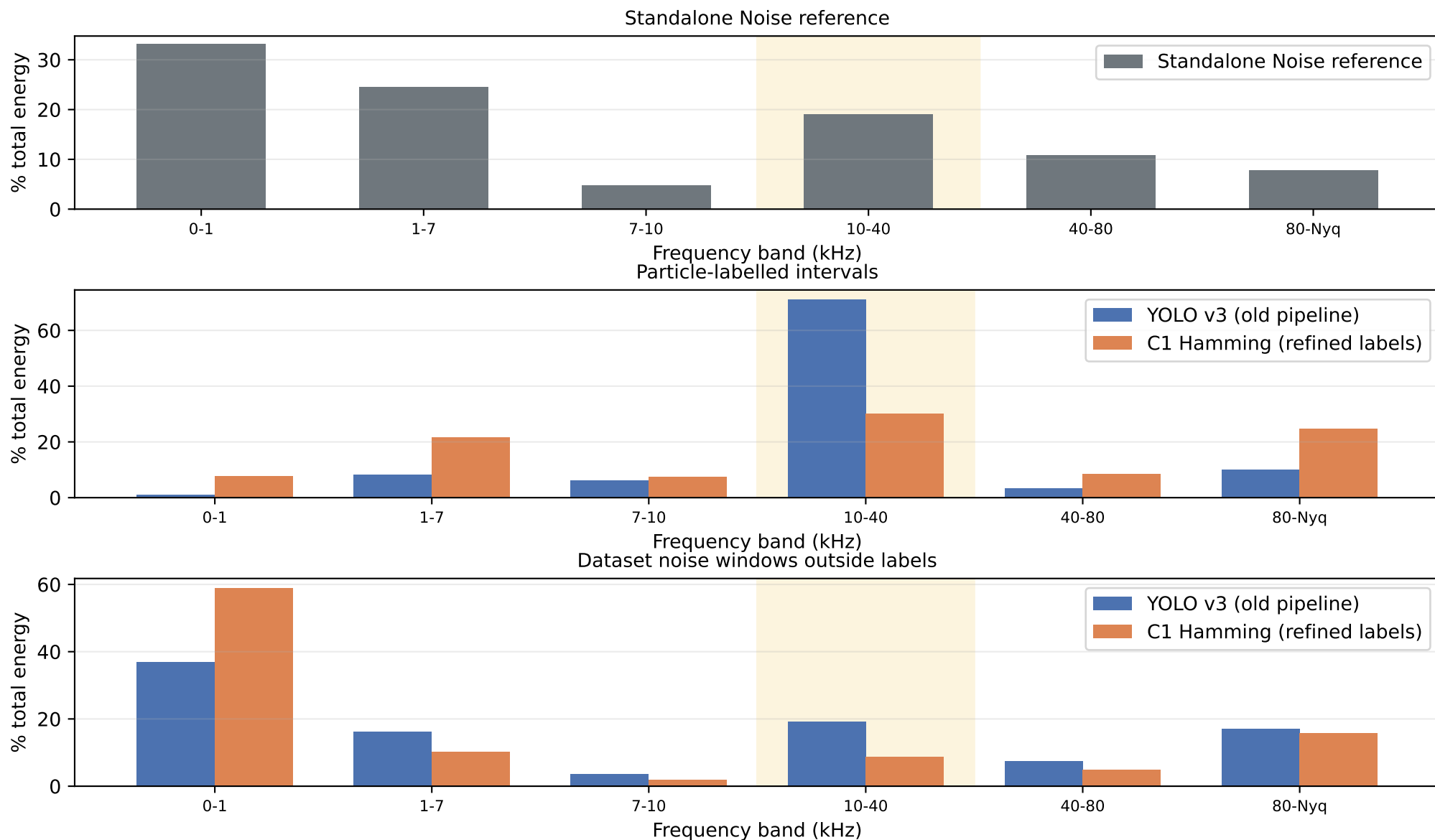
YOLO v3 (old pipeline) 10um: particle=72.2%, dataset-noise=19.2%, Noise reference=19.0%, dominant overlap=False

C1 Hamming (refined labels) 2um: particle=30.2%, dataset-noise=8.7%, Noise reference=19.0%, dominant overlap=False

C1 Hamming (refined labels) 4um: particle=51.5%, dataset-noise=8.9%, Noise reference=19.0%, dominant overlap=False

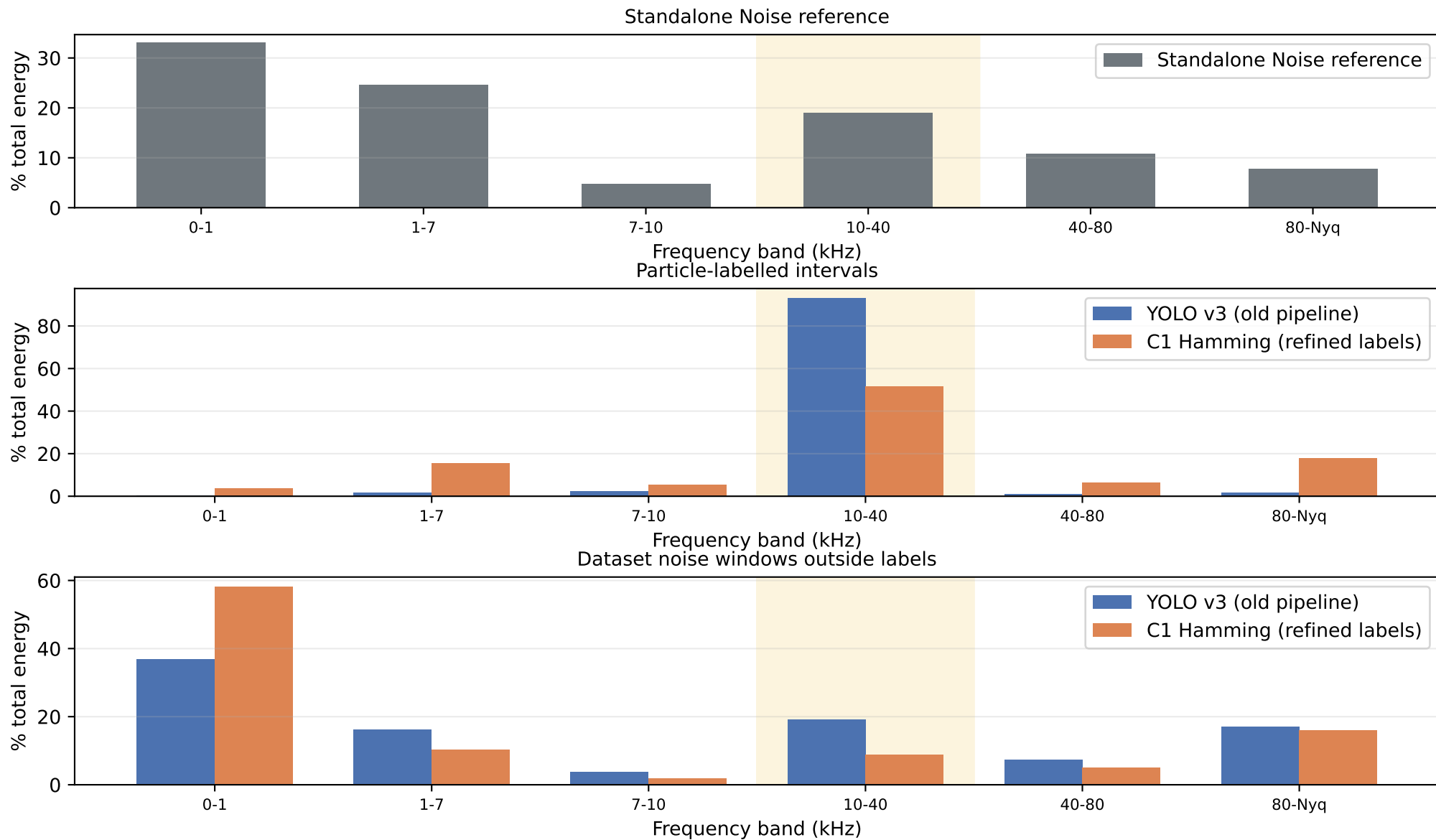
C1 Hamming (refined labels) 10um: particle=21.5%, dataset-noise=8.9%, Noise reference=19.0%, dominant overlap=False

2um - YOLO v3 (old pipeline) vs C1 Hamming (refined labels)



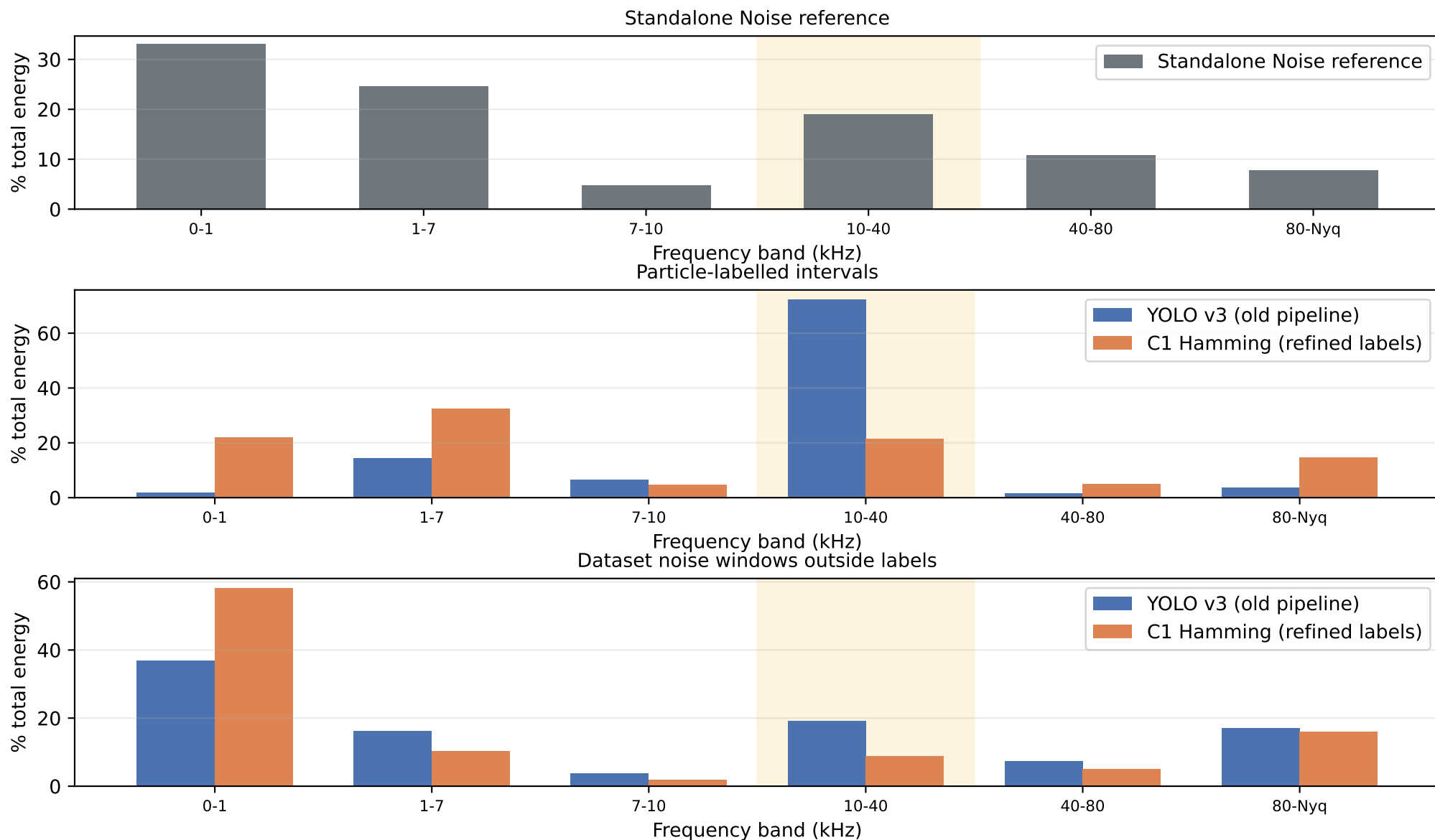
Dataset	Particle 10-40 kHz	Dataset-noise 10-40 kHz	Noise ref. 10-40 kHz	Mean label coverage
YOLO v3 (old pipeline)	71.0%	19.1%	19.0%	7.0%
C1 Hamming (refined labels)	30.2%	8.7%	19.0%	66.6%

4um - YOLO v3 (old pipeline) vs C1 Hamming (refined labels)



Dataset	Particle 10-40 kHz	Dataset-noise 10-40 kHz	Noise ref. 10-40 kHz	Mean label coverage
YOLO v3 (old pipeline)	93.0%	19.2%	19.0%	7.0%
C1 Hamming (refined labels)	51.5%	8.9%	19.0%	66.6%

10um - YOLO v3 (old pipeline) vs C1 Hamming (refined labels)



Dataset	Particle 10-40 kHz	Dataset-noise 10-40 kHz	Noise ref. 10-40 kHz	Mean label coverage
YOLO v3 (old pipeline)	72.2%	19.2%	19.0%	7.0%
C1 Hamming (refined labels)	21.5%	8.9%	19.0%	66.6%